Analogue electronics revision summary

ENGINEERING WITH MR KELLY

National 5 · Electronics & control · Class worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ recognise and draw the main components
- ☐ sort components into input and output transducers
- ☐ recall the formulae for series, parallel and voltage divider circuits
- ☐ describe how a transistor and relay work

I DO — Key facts

From the revision mind map

$V = IR$

Voltage = current × resistance (Ohm's law)

- V in volts, I in amps, R in ohms (Ω)
- Convert first: 4.7 kΩ = 4700 Ω, 30 mA = 0.03 A

	Series	Parallel
Resistance	$R_T = R_1 + R_2 + \ldots$	$1/R_T = 1/R_1 + 1/R_2 + \ldots$ (two: $R_1R_2 \div (R_1 + R_2)$)
Current	The same everywhere	Splits: $I_T = I_1 + I_2 + \ldots$
Voltage	Shared: $V_s = V_1 + V_2 + \ldots$	The same across every branch

$V_{out} = R_2 \div (R_1 + R_2) \times V_{supply}$

Voltage divider; also $V_1 \div V_2 = R_1 \div R_2$

- **Electrical energy** E_e = VIt; **electrical power** P = VI.
- **LDR**: light up, resistance down (LURD). **Thermistor**: temperature up, resistance down (TURD).
- **NPN transistor** (base, collector, emitter) switches on when about 0.7 V is applied to the base.
- A **relay** lets a small current in its coil switch a separate circuit through its contacts.

Remember

Check the data booklet for every formula.

YOU DO — Revision tasks

Use your notes if you get stuck.

1 Name the three connections of an NPN transistor, and state the voltage needed at the base to switch it on.

2 Explain why a relay is used in a control circuit.

3 State what happens to the resistance of an LDR as it gets darker, and of a thermistor as it gets hotter.

4 A 4 kΩ resistor and a 6 kΩ resistor form a divider on a 10 V supply. Calculate V_{out} across the 6 kΩ resistor.

5 Complete the table: draw each symbol, say whether it is an input or output transducer (or neither), and what it does.

Component	Draw the symbol	Input or output?	What it does
Battery			
Switch (SPST)			
Resistor			
Variable resistor			
LDR			
Thermistor			
Diode			
LED			
Lamp			
Buzzer			
Motor			

Check and reflect

Mark your answers with the answer sheet. Which “I can” statement above still needs work? Look back at the notes for that topic on the site.